* Continuity :-

Let, $f(x,y)$ be a function of x a y defined at (a,b) as well as in the neighbourhood of it. The function $f(x,y)$ is continuous at (a,b) if the following three conditions are satisfied.

i) $f(a,b)$ exists. i.e. $f(x,y)$ is defined at (a,b).

ii) $\lim_{(x,y)\to(a,b)} f(x,y)$ exists.

iii) $\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$

A function $f(x,y)$ is continuous in a domain if it is continuous at each point of that domain.

- If $f(x,y)$ and $g(x,y)$ are continuous at (a,b) then $f+g$, $f-g$, fg, f/g (provided $g \neq 0$) are continuous at (a,b).

Scanned with CamScanner

Examples:

① Show that $f(x,y) = x^2+2y$ is continuous at $(1,2)$.

In order to show the function $f(x,y) = x^2+2y$ is continuous, we will prove the following three points

1) $f(1,2)$ exists. i.e. $f(x,y)$ is defined at $(1,2)$.

$f(1,2) = 1^2+2(2) = 1+4 = 5$.

2) $\lim_{(x,y)\to(1,2)} f(x,y)$ exists.

$$L_1 = \lim_{(x,y)\to(1,2)} f(x,y) = \lim_{x\to 1}\left\{\lim_{y\to 2} x^2+2y\right\} = \lim_{x\to 1} x^2+4 = 5$$

$$L_2 = \lim_{y\to 2}\left\{\lim_{x\to 1} x^2+2y\right\} = \lim_{y\to 2}\{1+2y\} = 1+2(2) = 5$$

$\therefore \lim_{(x,y)\to(1,2)} f(x,y)$ exists.

3) $\lim_{(x,y)\to(1,2)} f(x,y) = f(1,2)$

$\therefore$ All the three conditions are satisfied.

$\therefore$ We can say that, $f(x,y)$ is continuous at $(1,2)$.

Note:

Every rational function is continuous.

② Determine the set of points at which the given function is continuous:

$$f(x,y) = \begin{cases} \dfrac{3x^2y}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0). \end{cases}$$

For $(x,y) \neq (0,0)$,

the function $f(x,y) = \dfrac{3x^2y}{x^2+y^2}$ is a rational function

$\therefore f(x,y)$ is continuous.

Scanned with CamScanner

For $(x,y) = (0,0)$,

$f(x,y) = 0$

$$\lim_{x\to 0}\left[\lim_{y\to 0} f(x,y)\right] = \lim_{x\to 0}\left[\lim_{y\to 0} \frac{3x^2y}{x^2+y^2}\right] = \lim_{x\to 0} 0 = 0.$$

$$\lim_{y\to 0}\left[\lim_{x\to 0} f(x,y)\right] = \lim_{y\to 0}\left[\lim_{x\to 0} \frac{3x^2y}{x^2+y^2}\right] = \lim_{y\to 0} 0 = 0$$

Putting $y = mx$ and taking limit $x \to 0$

$$\lim_{(x,y)\to(0,0)} \frac{3x^2y}{x^2+y^2} = \lim_{x\to 0} \frac{3x^2(mx)}{x^2+(mx)^2} = \lim_{x\to 0} \frac{3x^2m}{x^2+m^2x^2} = 0$$

Hence, the limit exists at $(0,0)$.

$\therefore \lim_{(x,y)\to(0,0)} f(x,y) = f(0,0)$

$\therefore$ 1) $\lim_{(x,y)\to(0,0)} f(x,y)$ exists.

2) $f(0,0)$ exists.

3) $\lim_{(x,y)\to(0,0)} f(x,y) = f(0,0)$

$\therefore$ All the three conditions are satisfied.

$\therefore$ The function is continuous at $(0,0)$.

Show that $f(x,y) = \begin{cases} \frac{2xy}{x^2+y^2}, & (x,y) \neq (0,0) \\ = 0, & (x,y) = (0,0) \end{cases}$

is continuous at every point except at the origin.

- For $(x,y) = (0,0)$

$$L_1 = \lim_{x\to 0}\left[\lim_{y\to 0} \frac{2xy}{x^2+y^2}\right] = \lim_{x\to 0}[0] = 0.$$

$$L_2 = \lim_{y\to 0}\left[\lim_{x\to 0} \frac{2xy}{x^2+y^2}\right] = \lim_{y\to 0}(0) = 0.$$

Putting $y = mx$ and taking limit $x \to 0$,

Scanned with CamScanner

$\lim_{(x,y)\to(0,0)} \frac{2xy}{x^2+y^2} = \lim_{x\to 0} \frac{2x(mx)}{x^2+(mx)^2} = \lim_{x\to 0} \frac{2x^2m}{x^2(1+m^2)}$

Now, L_3 depends on m and m is not fixed;
L_3 does not exist.

$\therefore$ $f(x,y)$ is not continuous at $(0,0)$.

– Let $f(x,y) = (a,b) \neq (0,0)$ be an arbitrary point in xy-plane, where a and b are real numbers.

1) $\lim_{(x,y)\to(a,b)} f(x,y) = \lim_{(x,y)\to(a,b)} \frac{2xy}{x^2+y^2} = \frac{2ab}{a^2+b^2}$

2) $f(a,b) = \frac{2ab}{a^2+b^2}$

3) $\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$

$\therefore$ All the three conditions are satisfied.
$\therefore$ $f(x,y)$ is continuous at (a,b)
$\therefore$ $f(x,y)$ is continuous at every point except at the origin.

Theorems on Limit:

If $\lim_{(x,y)\to(a,b)} f(x,y) = l$ and $\lim_{(x,y)\to(a,b)} g(x,y) = m$

i) $\lim_{(x,y)\to(a,b)} [f(x,y) + g(x,y)] = \lim_{(x,y)\to(a,b)} f(x,y) + \lim_{(x,y)\to(a,b)} g(x,y)$

ii) $\lim_{(x,y)\to(a,b)} [f(x,y) - g(x,y)] = \lim_{(x,y)\to(a,b)} f(x,y) - \lim_{(x,y)\to(a,b)} g(x,y)$

iii) $\lim_{(x,y)\to(a,b)} [f(x,y) \cdot g(x,y)] = \lim_{(x,y)\to(a,b)} f(x,y) \cdot \lim_{(x,y)\to(a,b)} g(x,y)$

iv) $\lim_{(x,y)\to(a,b)} \left[\frac{f(x,y)}{g(x,y)}\right] = \frac{\lim_{(x,y)\to(a,b)} f(x,y)}{\lim_{(x,y)\to(a,b)} g(x,y)} = \frac{l}{m}$, provided $m \neq 0$.

Scanned with CamScanner

Partial Derivatives:

A partial derivative of a function of several variables is the ordinary derivative w.r. to one of the variables, when all the remaining variables are kept constant.

- ~~The~~ partial consider a function $u = f(x, y)$.
  Here, u - dependent variable
  x, y - Independent variable.

- The partial derivative of $u = f(x, y)$ w.r. to x is the ordinary derivative of u w.r. to x keeping y constant.

- Denoted by $\frac{\partial u}{\partial x}$ or $\frac{\partial f}{\partial x}$ or u_x or f_x.

It is known as first order partial derivative of u w.r. to x,

$$\frac{\partial u}{\partial x} = \lim_{\Delta x \to 0} \left[\frac{f(x + \Delta x, y) - f(x, y)}{\Delta x} \right]$$

- Similarly, the partial derivative of $u = f(x, y)$ w.r. to y is the ordinary derivative of u w.r. to y treating x as constant.

- Denoted by $\frac{\partial u}{\partial y}$ or $\frac{\partial f}{\partial y}$ or u_y or u_x.

It is known as first order partial derivative of y w.r. to y

$$\frac{\partial u}{\partial y} = \lim_{\Delta y \to 0} \left[\frac{f(x, y + \Delta y) - f(x, y)}{\Delta y} \right]$$

e.g. $u = x + y$

$$\frac{\partial u}{\partial x} = 1 + 0 \quad , \quad \frac{\partial u}{\partial y} = 0 + 1$$

Scanned with CamScanner

Higher-Order Partial Derivatives:

Partial derivatives of higher order, of a function $u = f(x,y)$, are obtained by partial differentiation of first-order partial derivatives.

- Thus, if $u = f(x,y)$ then

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right), \quad \frac{\partial^2 u}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial x}\right)$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right), \quad \frac{\partial^2 u}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial y}\right).$$

are called second order partial derivatives

- Similarly, the other higher partial derivatives can also be obtained.

Clairaut's theorem:

If $u = f(x,y)$ possesses continuous second order partial derivatives $\frac{\partial^2 u}{\partial x \partial y}$ and $\frac{\partial^2 u}{\partial y \partial x}$ then

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$$

This is also called commutative property.

Note:- After this diff. of 1st of 1st at ex-(12)

standard rules for derivatives of sum, difference, product and quotient are also applicable for partial derivatives.

e.g. $u = x^2 + 2xy + y^2$

$$\frac{\partial u}{\partial x} = 2x + 2y, \quad \frac{\partial u}{\partial y} = 2x + 2y$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right) = \frac{\partial}{\partial x}(2x+2y) = 2$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right) = \frac{\partial}{\partial y}(2x+2y) = 2.$$

Scanned with CamScanner

Examples:

(1) If $u = (1-2xy+y^2)^{-1/2}$ then show that $x\frac{\partial u}{\partial x} - y\frac{\partial u}{\partial y} = u^3y^2$.

$$u = (1-2xy+y^2)^{-1/2}$$

$$\frac{\partial u}{\partial x} = -\tfrac{1}{2}(1-2xy+y^2)^{-\frac{1}{2}-1}(-2y) = y(1-2xy+y^2)^{-3/2}$$

$$\frac{\partial u}{\partial y} = -\tfrac{1}{2}(1-2xy+y^2)^{-\frac{1}{2}-1}(-2x+2y) = -\frac{(2y-2x)}{2}(1-2xy+y^2)^{-\frac{3}{2}}$$

$$x\frac{\partial u}{\partial x} - y\cdot\frac{\partial u}{\partial y} = xy(1-2xy+y^2)^{-3/2} + \frac{y}{2}(2y-2x)(1-2xy+y^2)^{-\frac{3}{2}}$$

$$= (1-2xy+y^2)^{-3/2}\left[xy + y(y-x)\right]$$

$$= (1-2xy+y^2)^{-3/2}\left[xy + y^2 - xy\right]$$

$$= \left[(1-2xy+y^2)^{-1/2}\right]^3 y^2$$

$$= u^3y^2.$$

(2) If $u = \log(\tan x + \tan y + \tan z)$ then show that $\sin 2x\frac{\partial u}{\partial x} + \sin 2y\frac{\partial u}{\partial y} + \sin 2z\frac{\partial u}{\partial z} = 2$.

$$u = \log(\tan x + \tan y + \tan z)$$

$$\frac{\partial u}{\partial x} = \frac{1}{\tan x + \tan y + \tan z}\cdot\sec^2 x$$

$$\frac{\partial u}{\partial y} = \frac{1}{\tan x + \tan y + \tan z}\sec^2 y$$

$$\frac{\partial u}{\partial z} = \frac{1}{\tan x + \tan y + \tan z}\cdot\sec^2 z.$$

$$\sin 2x\frac{\partial u}{\partial x} + \sin 2y\frac{\partial u}{\partial y} + \sin 2z\frac{\partial u}{\partial z} =$$

$$\sin 2x\,\frac{\sec^2 x}{\tan x + \tan y + \tan z} + \sin 2y\cdot\frac{\sec^2 y}{\tan x + \tan y + \tan z} +$$

$$\sin 2z\cdot\frac{\sec^2 z}{\tan x + \tan y + \tan z}.$$

Scanned with CamScanner

$$= \frac{2\sin x \cdot \cos x \frac{1}{\cos^2 x} + 2\sin y \cos y \frac{1}{\cos^2 y} + 2\sin z \cdot \cos z \cdot \frac{1}{\cos^2 z}}{\tan x + \tan y + \tan z}$$

$$= \frac{2(\tan x + \tan y + \tan z)}{\tan x + \tan y + \tan z}$$

$$= 2.$$

(3) Find the value of n for which $u = kt^{-1/2} e^{-\left(\frac{x^2}{na^2t}\right)}$ satisfies the equation $\frac{\partial u}{\partial t} = a^2 \frac{\partial^2 u}{\partial x^2}$.

$$u = kt^{-1/2} e^{-\left(\frac{x^2}{na^2t}\right)}$$

$$\frac{du}{dt} = k\cdot\left(-\frac{1}{2}\right)t^{-1/2-1} e^{-\left(\frac{x^2}{na^2t}\right)} + kt^{-1/2} e^{-\left(\frac{x^2}{na^2t}\right)} \left\{-\left(\frac{x^2(-1)t^{-2}}{na^2}\right)\right.$$

$$= -\frac{1}{2}kt^{-3/2} e^{-\left(\frac{x^2}{na^2t}\right)} + kt^{-1/2} e^{-\left(\frac{x^2}{na^2t}\right)} \left(\frac{x^2}{na^2t^2}\right)$$

$$= -\frac{1}{2}t^{-1}u + u\left(\frac{x^2}{na^2t^2}\right)$$

$$= u\left(\frac{x^2}{na^2t^2} - \frac{1}{2t}\right) \quad \text{— (1)}$$

$$\frac{du}{dx} = kt^{-1/2} e^{-\left(\frac{x^2}{na^2t}\right)} \cdot \left(\frac{-2x}{na^2t}\right)$$

$$\frac{d^2u}{dx^2} = kt^{-1/2} e^{-\left(\frac{x^2}{na^2t}\right)} \left(\frac{-2x}{na^2t}\right)\left(\frac{-2x}{na^2t}\right) + kt^{-1/2} e^{-\left(\frac{x^2}{na^2t}\right)} \left(\frac{-2}{na^2t}\right.$$

$$= \frac{kt^{-1/2}}{na^2t} e^{-\left(\frac{x^2}{na^2t}\right)} \left[\frac{4x^2}{na^2t} - 2\right]$$

$$a^2\frac{\partial^2 u}{\partial x^2} = kt^{-1/2} u \left(\frac{4x^2}{n^2a^2t^2} - \frac{2}{nt}\right) \quad \text{— (2)}$$

Scanned with CamScanner

From (1) & (2),

$$u\left(\frac{x^2}{na^2t^2} - \frac{1}{2t}\right) = u\left(\frac{4x^2}{n^2a^2t^2} - \frac{2}{nt}\right)$$

$$\therefore \frac{x^2}{na^2t^2} - \frac{1}{2t} = \frac{4x^2}{n^2a^2t^2} - \frac{2}{nt}$$

$$\therefore 2nx^2 - na^2t^2 = 2na^2t^2 ($$

$$\therefore (2x^2 - na^2t)na^2t^2 = 2na^2t^2(4x^2 - 2nta^2)$$

$$\therefore (2x^2 - na^2t)n = 4(2x^2 - na^2t)$$

$$\therefore \boxed{n = 4}$$

4) If $z^3 - zx - y = 0$, find $\frac{\partial^2 z}{\partial x \partial y}$.

$$z^3 - zx - y = 0 \quad — (1)$$

To find $\frac{\partial^2 z}{\partial x \partial y}$, we have to find $\frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right)$

To find first $\frac{\partial z}{\partial y}$.

Diff. eq^n (1) w.r. to y,

$$3z^2\frac{\partial z}{\partial y} - \frac{\partial z}{\partial y}x - 1 = 0$$

$$\frac{\partial z}{\partial y}(3z^2 - x) - 1 = 0$$

$$\therefore \frac{\partial z}{\partial y}(3z^2 - x) = 1$$

$$\therefore \frac{\partial z}{\partial y} = \frac{1}{3z^2 - x}.$$

Differentiate eq^n (1), partially w.r. to x,

$$3z^2\frac{\partial z}{\partial x} - \frac{\partial z}{\partial x}x - z = 0$$

$$\therefore \frac{\partial z}{\partial x}(3z^2 - x) - z = 0 \Rightarrow \frac{\partial z}{\partial x} = \frac{z}{3z^2 - x}.$$

Scanned with CamScanner

Now, Diff. $\frac{\partial z}{\partial y}$ partially w.r. to x,

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial}{\partial x}\left(\frac{1}{3z^2 - x}\right)$$

$$= \frac{\partial}{\partial x}(3z^2 - x)^{-1}$$

$$= -(3z^2 - x)^{-2}\left(6z\frac{\partial z}{\partial x} - 1\right)$$

$$= -\frac{1}{(3z^2 - x)^2}\left[6z \cdot \frac{z}{(3z^2 - x)} - 1\right]$$

$$= -\frac{1}{(3z^2 - x)^2}\left[\frac{6z^2}{(3z^2 - x)} - 1\right]$$

$$= -\left[\frac{6z^2}{(3z^2 - x)^3} - \frac{1}{(3z^2 - x)^2}\right]$$

$$= -\left[\frac{6z^2 - 3z^2 + x}{(3z^2 - x)^3}\right]$$

$$= -\frac{3z^2 + x}{(3z^2 - x)^3}$$

(5) If $u = e^{xyz}$, show that $\frac{\partial^3 u}{\partial x \partial y \partial z} = (1 + 3xyz + x^2y^2z^2)e^{xyz}$

Here, $u = e^{xyz}$

$$\frac{\partial^3 u}{\partial x \partial y \partial z} = \frac{\partial^2}{\partial x \partial y}\left(\frac{\partial u}{\partial z}\right) = \frac{\partial}{\partial x}\left(\frac{\partial^2 u}{\partial x \partial z}\right)$$

Diff. u partially w.r. to z,

$$\frac{\partial u}{\partial z} = e^{xyz} \cdot xy$$

Scanned with CamScanner

$$\frac{\partial}{\partial y}\left(\frac{\partial u}{\partial z}\right) = \frac{\partial}{\partial y}\left(e^{xyz} \cdot xy\right)$$

$$= e^{xyz}(xz)(xy) + e^{xyz} \cdot x$$

$$\frac{\partial^2 u}{\partial y \partial z} = x^2yz\, e^{xyz} + x e^{xyz}$$

Differentiate $\frac{\partial^2 u}{\partial y \partial z}$ partially w.r. to x,

$$\frac{\partial}{\partial x}\left(\frac{\partial^2 u}{\partial y \partial z}\right) = \frac{\partial}{\partial x}\left(x^2yz\, e^{xyz} + x e^{xyz}\right)$$

$$= 2xyz\, e^{xyz} + x^2yz \cdot e^{xyz} \cdot yz + e^{xyz} + x \cdot e^{xyz} \cdot yz$$

$$= 3xyz\, e^{xyz} + x^2y^2z^2 e^{xyz} + e^{xyz}$$

$$= e^{xyz}(1 + 3xyz + x^2y^2z^2).$$

3) If $z = x + y^x$, prove that $\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$.

Here, $z = x + y^x$

Differentiate partially w.r. to x,

$$\frac{\partial z}{\partial x} = 1 + y^x \cdot \log y$$

Differentiate partially w.r. to y,

$$\frac{\partial z}{\partial y} = xy^{x-1}$$

Differentiate $\frac{\partial z}{\partial x}$ partially w.r. to y

$$\frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial}{\partial y}\left(1 + y^x \cdot \log y\right) = xy^{x-1}(\log y) + y^x \cdot \frac{1}{y}$$

$$= xy^{x-1}(\log y) + y^{x-1}$$

$$\frac{\partial^2 z}{\partial y \partial x} = y^{x-1}(1 + x \cdot \log y) \quad -(1$$

Scanned with CamScanner

Differentiating $\frac{\partial z}{\partial y}$ partially w.r.to x,

$$\frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x}\left(x y^{x-1}\right)$$

$$= 1 \cdot y^{x-1} + x \cdot y^{x-1} \cdot \log y$$

$$= y^{x-1}(1 + x \log y) \quad -(2)$$

$\therefore$ From eq^{n} (1) & (2),

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}.$$

(7) If $u = \log(x^3+y^3+z^3-3xyz)$, prove that

$$\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)^2 u = -\frac{9}{(x+y+z)^2}$$

$$\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)^2 u = \left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right) u$$

$$= \left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)\left(\frac{\partial u}{\partial x}+\frac{\partial u}{\partial y}+\frac{\partial u}{\partial z}\right)$$

$$= \left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right) v$$

where $v = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}$.

Now, $u = \log(x^3+y^3+z^3-3xyz)$

$$\frac{\partial u}{\partial x} = \frac{1}{x^3+y^3+z^3-3xyz}(3x^2-3yz)$$

$$\frac{\partial u}{\partial y} = \frac{1}{x^3+y^3+z^3-3xyz}(3y^2-3xz)$$

$$\frac{\partial u}{\partial z} = \frac{1}{x^3+y^3+z^3-3xyz}(3z^2-3xy)$$

Now, $v = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}$.

Scanned with CamScanner

$$\therefore u = \frac{3x^2 - 3yz + 3y^2 - 3xz + 3z^2 - 3xy}{x^3+y^3+z^3-3xyz}$$

$$= \frac{3(x^2+y^2+z^2-xy-yz-zx)}{x^3+y^3+z^3-3xyz} \cdot \frac{x+y+z}{x+y+z}$$

$$= \frac{3(x^3+y^3+z^3-3xyz)}{x^3+y^3+z^3-3xyz(x+y+z)}$$

$$= \frac{3}{x+y+z}$$

$(a^3+b^3) = (a+b)(a^2-ab+b^2)$

$a^3+b^3+c^3 = (a+b+c)(a^2+b^2+c^2-ab-bc-ca)$

Now, $\left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)^2 u = \left(\frac{\partial}{\partial x}+\frac{\partial}{\partial y}+\frac{\partial}{\partial z}\right)\left(\frac{3}{x+y+z}\right)$

$$= -\frac{3}{(x+y+z)^2} - \frac{3}{(x+y+z)^2} - \frac{3}{(x+y+z)^2}$$

$$= -\frac{9}{(x+y+z)^2}$$

) If $x = r\cos\theta$, $y = r\sin\theta$, show that $\left(\frac{\partial r}{\partial x}\right)^2 + \left(\frac{\partial r}{\partial y}\right)^2 = 1$.

Here, $x = r\cos\theta$, $y = r\sin\theta$

$\therefore x^2 = r^2\cos^2\theta$, $y^2 = r^2\sin^2\theta$

$\therefore x^2+y^2 = r^2(\cos^2\theta + \sin^2\theta)$

$\therefore x^2+y^2 = r^2$ — (1)

To find $\frac{\partial r}{\partial x}$, diff. eq^n (1) w.r. to x,

$2x = 2r\frac{\partial r}{\partial x}$

$\therefore \frac{\partial r}{\partial x} = \frac{2x}{2r} = \frac{x}{r}$.

Diff. eq^n (1) partially w.r. to y,

Scanned with CamScanner

$2y = 2r\frac{\partial r}{\partial y}$

$\therefore \frac{\partial r}{\partial y} = \frac{2y}{2r} = \frac{y}{r}.$

$\therefore \left(\frac{\partial r}{\partial x}\right)^2 + \left(\frac{\partial r}{\partial y}\right)^2 = \frac{x^2}{r^2} + \frac{y^2}{r^2} = \frac{x^2+y^2}{r^2} = \frac{r^2}{r^2} = 1.$

(9) If $x = \cos\theta - r\sin\theta$, $y = \sin\theta + r\cos\theta$, show that

i) $\frac{\partial r}{\partial x} = \frac{x}{r}$ ii) $\frac{\partial \theta}{\partial x} = -\frac{\cos\theta}{r}$.

Here $x = \cos\theta - r\sin\theta$ — (1)

$y = \sin\theta + r\cos\theta$ — (2)

prove i) as above example.

ii) Multiply eq^n (1) by $\cos\theta$ & eq^n (2) by $\sin\theta$

$x\cos\theta = \cos^2\theta - r\sin\theta\cos\theta$ — (3)

$y\sin\theta = \sin^2\theta + r\sin\theta\cos\theta$ — (4)

Adding eq^n (3) & (4),

$x\cos\theta + y\sin\theta = 1$

dividing ~~Multiplying~~ by $\sin\theta$,

$x\cot\theta + y = \operatorname{cosec}\theta$ — (5)

Diff. eq^n (5) partially w.r. to x

$\cot\theta + x(-\operatorname{cosec}^2\theta)\frac{\partial\theta}{\partial x} = -\cot\theta\cdot\operatorname{cosec}\theta\frac{\partial\theta}{\partial x}.$

$\cot\theta + \cot\theta\cdot\operatorname{cosec}\theta\frac{\partial\theta}{\partial x} = x\operatorname{cosec}^2\theta\frac{\partial\theta}{\partial x}$

$= (\cos\theta - r\sin\theta)\operatorname{cosec}^2\theta\frac{\partial\theta}{\partial x}$

$= \cos\theta\cdot\operatorname{cosec}^2\theta\frac{\partial\theta}{\partial x} - r\sin\theta\operatorname{cosec}^2$

$= \cot\theta\cdot\operatorname{cosec}\theta\frac{\partial\theta}{\partial x} - r\operatorname{cosec}\theta\frac{\partial\theta}{\partial x}$

Scanned with CamScanner

$$\cot\theta = -r\,\mathrm{cosec}\,\theta\,\frac{\partial\theta}{\partial x}$$

$$\therefore \frac{\partial\theta}{\partial x} = -\frac{\cot\theta}{r\cdot\mathrm{cosec}\,\theta}$$

$$= -\frac{\cos\theta}{r}$$

5) Show that $\frac{\partial x}{\partial r} = \frac{1}{r}\frac{\partial y}{\partial\theta}$ and $\frac{\partial y}{\partial r} = -\frac{1}{r}\frac{\partial x}{\partial\theta}$ and hence show that $\frac{\partial^2 x}{\partial r^2} + \frac{1}{r}\frac{\partial x}{\partial r} + \frac{1}{r^2}\frac{\partial^2 x}{\partial\theta^2} = 0$ if $x = e^{r\cos\theta}\cos(r\sin\theta)$ and $y = e^{r\cos\theta}\sin(r\sin\theta)$.

Here, $x = e^{r\cos\theta}\cos(r\sin\theta)$ & $y = e^{r\cos\theta}\sin(r\sin\theta)$

Diff., x partially w.r. to r,

$$\frac{\partial x}{\partial r} = e^{r\cos\theta}(\cos\theta)\cos(r\sin\theta) + e^{r\cos\theta}[-\sin(r\sin\theta)](\sin\theta)$$

$$= e^{r\cos\theta}[\cos\theta\cos(r\sin\theta) - \sin\theta\sin(r\sin\theta)]$$

$$= e^{r\cos\theta}\cos(\theta + r\sin\theta) \quad \text{—— (1)}$$

Diff. y partially w.r. to θ,

$$\frac{\partial y}{\partial\theta} = e^{r\cos\theta}\cdot r(-\sin\theta)\sin(r\sin\theta) + e^{r\cos\theta}\cos(r\sin\theta)(r\cos\theta)$$

$$= re^{r\cos\theta}[\cos(r\sin\theta)(\cos\theta) - \sin(r\sin\theta)\sin\theta]$$

$$= re^{r\cos\theta}\cos(\theta + r\sin\theta) \quad \text{—— (2)}$$

From (1) & (2)

$$\frac{\partial x}{\partial r} = \frac{1}{r}\frac{\partial y}{\partial\theta} \quad \text{—— (3)}$$

Now, Diff. x partially w.r. to θ,

$$\frac{\partial x}{\partial\theta} = e^{r\cos\theta}(-r\sin\theta)\cos(r\sin\theta) + e^{r\cos\theta}[-\sin(r\sin\theta)(r\cos\theta)]$$

$$= -re^{r\cos\theta}[\sin\theta\cdot\cos(r\sin\theta) + \sin(r\sin\theta)\cos\theta]$$

$$= -re^{r\cos\theta}\sin(\theta + r\sin\theta) \quad \text{—— (4)}$$

Scanned with CamScanner

Diff. y partially w.r.to r,

$$\frac{dy}{dr} = e^{r\cos\alpha}(\cos\alpha)\sin(r\sin\alpha) + e^{r\cos\alpha}[\cos(r\sin\alpha)(\sin\alpha)]$$

$$= e^{r\cos\alpha}[\cos\alpha\cdot\sin(r\sin\alpha) + \cos(r\sin\alpha)(\sin\alpha)]$$

$$= e^{r\cos\alpha}\sin(r\sin\alpha+\alpha) \quad \text{——} (5)$$

From (3) & (4),

$$\frac{\partial x}{\partial\alpha} = -r\frac{dy}{dr} \quad \text{——} (6)$$

$$\therefore \frac{dy}{dr} = -\frac{1}{r}\frac{\partial x}{\partial\alpha}$$

Now, diff. eqⁿ (3) partially w.r. to r,

$$\frac{d^2x}{dr^2} = \frac{\partial}{\partial r}\left(\frac{1}{r}\frac{dy}{d\alpha}\right)$$

$$= -\frac{1}{r^2}\frac{dy}{d\alpha} + \frac{1}{r}\frac{d^2y}{dr\,d\alpha}$$

For $\frac{\partial^2 x}{\partial\alpha^2}$, diff, eqⁿ (6) partially w.r.to α

$$\frac{\partial^2 x}{\partial\alpha^2} = \frac{\partial}{\partial\alpha}\left(-r\frac{dy}{dr}\right)$$

$$= -r\frac{\partial^2 y}{\partial\alpha\,dr}$$

$$= -r\frac{\partial^2 y}{dr\,d\alpha}$$

Now, $\frac{\partial^2 x}{dr^2} + \frac{1}{r}\frac{\partial x}{\partial r} + \frac{1}{r^2}\frac{\partial^2 x}{\partial\alpha^2} = -\frac{1}{r^2}\frac{dy}{d\alpha} + \frac{1}{r^2}\frac{dy}{d\alpha} - \frac{1}{r}\frac{\partial^2 y}{dr\,d\alpha} + \frac{1}{r}\frac{\partial^2 y}{dr\,d\alpha}$

$$= 0.$$

Scanned with CamScanner

(11) If $u = \phi(x+ky) + \psi(x-ky)$, show that $\frac{\partial^2 u}{\partial y^2} = k^2 \frac{\partial^2 u}{\partial x^2}$.

Here, $u = \phi(x+ky) + \psi(x-ky)$

Diff. u partially w.r. to x,

$$\frac{\partial u}{\partial x} = \phi'(x+ky) + \psi'(x-ky)$$

$$\therefore \frac{\partial^2 u}{\partial x^2} = \phi''(x+ky) + \psi''(x-ky) \quad -(1)$$

$$\frac{\partial u}{\partial y} = \phi'(x+ky)\, k + \psi'(x-ky)\,(-k)$$

$$= k\,\phi'(x+ky) - k\,\psi'(x-ky).$$

$$\frac{\partial^2 u}{\partial y^2} = k^2 \phi''(x+ky) + k^2 \psi''(x-ky)$$

$$= k^2 \left[\phi''(x+ky) + \psi''(x-ky)\right] \quad -(2)$$

$\therefore$ From (1) & (2), $\frac{\partial^2 u}{\partial y^2} = k^2 \frac{\partial^2 u}{\partial x^2}$.

(2) If $u = f(r)$ and $r^2 = x^2+y^2+z^2$, prove that

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = f''(r) + \frac{2}{r} f'(r).$$

Here, $u = f(r)$

Diff. u partially w.r. to x,

$$\frac{\partial u}{\partial x} = f'(r) \frac{\partial r}{\partial x} \quad -(1)$$

But $r^2 = x^2+y^2+z^2$

$$2r \cdot \frac{\partial r}{\partial x} = 2x$$

$$\frac{\partial r}{\partial x} = \frac{x}{r}$$

From eq^n (1),

$$\frac{\partial u}{\partial x} = f'(r) \frac{x}{r}$$

$$\frac{\partial^2 u}{\partial x^2} = f''(r) \frac{\partial r}{\partial x} \frac{x}{r} + f'(r) \cdot \frac{1}{r} + f'(r) \left(-\frac{x}{r^2}\right) \frac{\partial r}{\partial x}$$

Scanned with CamScanner

$$\therefore \frac{\partial^2 u}{\partial x^2} = f''(r)\frac{x}{r}\frac{x}{r} + \frac{f'(r)}{r} - \frac{x^2}{r^3} f'(r)$$

$$= f''(r)\frac{x^2}{r^2} + \frac{f'(r)}{r} - \frac{x^2}{r^3} f'(r). \quad \text{—} (2)$$

In the same way,

$$\frac{\partial^2 u}{\partial y^2} = f''(r)\frac{y^2}{r^2} + \frac{f'(r)}{r} - \frac{y^2}{r^3} f'(r) \quad \text{——} (3)$$

$$\frac{\partial^2 u}{\partial z^2} = f''(r)\frac{z^2}{r^2} + \frac{f'(r)}{r} - \frac{z^2}{r^3} f'(r) \quad \text{——} (4)$$

(2) + (3) + (4)

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \frac{f''(r)}{r^2}(x^2+y^2+z^2) + 3\frac{f'(r)}{r} - \frac{(x^2+y^2+z^2)}{r^3} f'(r)$$

$$= \frac{f''(r)}{r^2} \cdot r^2 + 3\frac{f'(r)}{r} - \frac{r^2}{r^3} f'(r)$$

$$= f''(r) + 2\frac{f'(r)}{r}.$$

(13) If $x^x y^y z^z = c$, show that at $x = y = z$, $\frac{\partial^2 z}{\partial x \partial y} = -(x \log e)^{-1}$

Here $x^x y^y z^z = c$

Taking log both the sides

$\log x^x + \log y^y + \log z^z = \log c$

$x \log x + y \log y + z \log z = \log c \quad \text{—} (1)$

Diff. (1) partially w.r.to y

$$0 + 1 \cdot \log y + y \cdot \frac{1}{y} + \frac{\partial z}{\partial y} \log z + z \cdot \frac{1}{z} \frac{\partial z}{\partial y} = 0$$

$$\log y + 1 + \frac{\partial z}{\partial y}(\log z + 1) = 0$$

$$\frac{\partial z}{\partial y}(1 + \log z) = -(1 + \log y)$$

$$\therefore \frac{\partial z}{\partial y} = -\frac{(1+\log y)}{1+\log z}$$

Scanned with CamScanner

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right)$$

$$= \frac{\partial}{\partial x}\left[-\frac{(1+\log y)}{1+\log z}\right]$$

$$= -(1+\log y)\ \frac{\partial}{\partial x}(1+\log z)^{-1}$$

$$= -(1+\log y)\ -(1+\log z)^{-2}\cdot\left(0+\frac{1}{z}\frac{\partial z}{\partial x}\right) \quad \text{—— (2)}$$

Now, For $\frac{\partial z}{\partial x}$: $1\cdot\log x + x\cdot\frac{1}{x} + 0 + \frac{\partial z}{\partial x}\log z + z\cdot\frac{1}{z}\frac{\partial z}{\partial x} = 0$

$$\log x + 1 + (\log z + 1)\frac{\partial z}{\partial x} = 0$$

$$\therefore \frac{\partial z}{\partial x} = -\frac{(1+\log x)}{(1+\log z)}$$

Now, From (2),

$$\frac{\partial^2 z}{\partial x \partial y} = 0 + \frac{(1+\log y)}{(1+\log z)^2}\cdot\left[-\frac{(1+\log x)}{z(1+\log z)}\right]$$

$$= -\frac{(1+\log y)(1+\log x)}{(1+\log z)^3 \cdot z}$$

At $x = y = z$,

$$\frac{\partial^2 z}{\partial x \partial y} = -\frac{(1+\log x)(1+\log x)}{x(1+\log x)^3}$$

$$= -\frac{1}{x(1+\log x)}$$

$$= -[x(1+\log x)]^{-1}$$

$$= -[x(\log e + \log x)]^{-1}$$

$$= -[x(\log ex)]^{-1}$$

Scanned with CamScanner